PaperFuture

Forecasting future research through trajectory-guided technical leaps

Domains: interpretable multimodal self-evolution / Agent security / AI4EDA

All model inference via APIs

A Historical preparation · Reused for each domain, subfield and cutoff T

1 Literature collection & selection

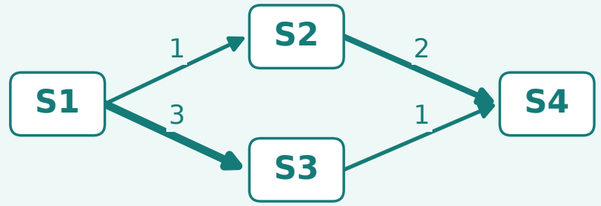
Collect metadata and resolve paper versions

Lexical + dense retrieval → rerank for coverage

Selected data: versioned papers + full-text evidence

2 Weighted directed trajectory graph

Nodes: stages, papers, mechanisms
Edges: technical inheritance
Width: technical change
Branches, merges, bottlenecks



Edge weight is change magnitude, not success probability

Global stage graph + key paths + source evidence

B Adaptive exploration · One candidate per round · At most four rounds in total

3 LLM leap proposal

Locate a bottleneck in the trajectory

Specify the technical change

Choose and justify leap magnitude

Idea + explicit assumptions

4 World model: simulate & discuss

Conditional simulation
Outcomes, failures and next bottlenecks

Independent review
Evidence-backed objections; explicit unknowns

Reconcile and assess
Preserve novelty; assess value and feasibility

5 Adaptive decision

Good enough?
Stalled?
Round 4?

Fixed decision rules
No new candidate generation

Stop

Continue

Forecast feedback + retained assumptions + next bottleneck → next leap (t + 1)

6 Idea & assessment

Main idea
Problem, mechanism, novelty

Value & feasibility
Conditions and uncertainty

Validation plan
Minimal test and sources

Best assessed candidate

Solid edge width: historical change Dashed loop: simulated feedback; never promoted to historical fact

Proposed framework · Simulated assessments are not experimental results